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Faculty of Natural and Applied Sciences

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SEN 481: SOFTWARE ENGINEERING ROBOTICS

AUTONOMOUS FIRE AND SMOKE DETECTION SPIDER ROBOT (QUADRUPED)

Technical Documentation and Project Report

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1. INTRODUCTION

1.1 Project Overview

The Autonomous Fire and Smoke Detection Spider Robot is a 4-legged (quadruped) biologically-inspired robot designed for autonomous fire and smoke detection in environments where early warning systems are critical. This robot combines advanced sensor technology with intelligent spider-like locomotion to navigate spaces and detect potential fire hazards before they escalate into dangerous situations.

The project integrates multiple sensing technologies including MQ-2 gas/smoke sensors and IR flame detection sensors, controlled by an Arduino Nano microcontroller. The quadruped spider design provides stable movement across varied terrain, making it suitable for deployment in warehouses, factories, server rooms, and other facilities requiring continuous fire monitoring.

1.2 Problem Statement

Fire incidents continue to pose significant threats to lives and property worldwide. Traditional fire detection systems are stationary and limited in their coverage area. There is a growing need for mobile, intelligent fire detection systems that can actively patrol areas, detect early signs of fire or smoke, and alert authorities before fires spread. This project addresses this need by developing an autonomous mobile spider robot capable of real-time fire and smoke detection.

1.3 Project Objectives

- Design and construct a functional quadruped spider robot platform with stable locomotion
- Integrate smoke detection using MQ-2 gas sensor module
- Implement flame detection using IR flame sensor with LM393 comparator
- Enable obstacle avoidance using HC-SR04 ultrasonic sensors
- Implement wireless communication for remote monitoring via Bluetooth (HC-05/HC-06)
- Develop alert mechanisms for fire/smoke detection events

1.4 Scope and Limitations

The scope of this project includes the design, assembly, programming, and testing of a quadruped spider robot with integrated fire and smoke detection capabilities. The robot operates autonomously in controlled indoor environments with Bluetooth connectivity for real-time data transmission. Limitations include operational range bounded by Bluetooth communication distance (approximately 10 meters), battery life constraints, and sensor sensitivity ranges.

2. SYSTEM ARCHITECTURE

2.1 System Block Diagram

The system architecture consists of four main subsystems: Power Management, Central Processing Unit, Sensor Array, and Actuation System. These subsystems work together to enable autonomous navigation, environmental sensing, and alert generation.

```
[18650 Battery Pack] --> [LM2596 Buck Converter] --> [Arduino Nano] --> [Servo Shield]
                                                          |                  |
                                          [Sensors Array]          [12x SG90 Servos]
                                                                (4 Legs x 3 Joints)
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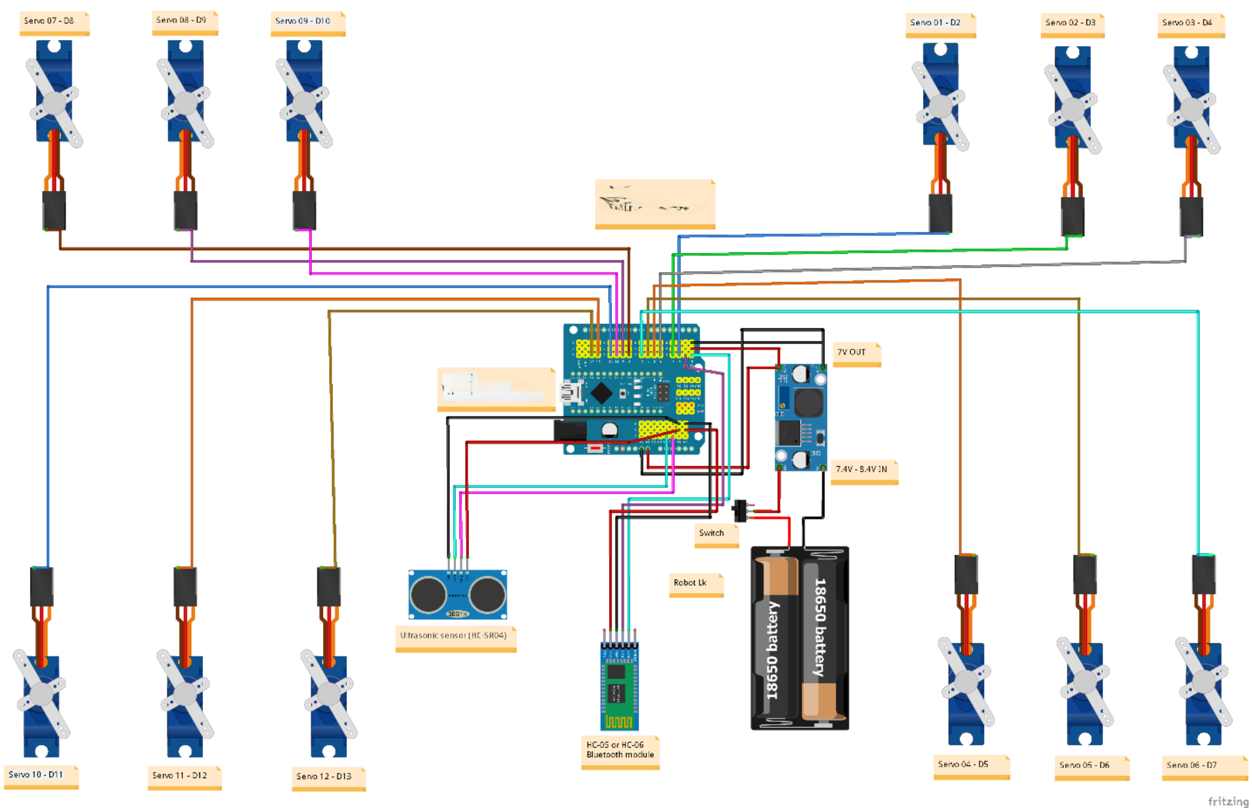

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2.2 Functional Description

Power Subsystem: Two 18650 lithium-ion batteries connected in series provide 7.4V nominal voltage. The LM2596 buck converter regulates this to stable 5V-6V output suitable for the Arduino Nano and servo motors.
Processing Subsystem: The Arduino Nano V3.0 serves as the central processing unit, running the control algorithms, sensor data processing, and communication protocols.

Sensing Subsystem: Multiple sensors provide environmental awareness including gas/smoke detection (MQ-2), flame detection (IR sensor), distance measurement (HC-SR04), and wireless communication (HC-05/HC-06 Bluetooth).
Actuation Subsystem: Twelve SG90 micro servo motors (three per leg for 4 legs) enable quadruped spider locomotion for stable movement.

3. COMPONENT SPECIFICATIONS AND DATASHEETS

3.1 Arduino Nano V3.0 (ATmega328P)

The Arduino Nano is a compact, breadboard-friendly microcontroller board based on the ATmega328P. It serves as the brain of the robot, controlling all subsystems.

Parameter	Value
Microcontroller	ATmega328P
Operating Voltage	5V
Input Voltage (recommended)	7-12V
Digital I/O Pins	14 (6 provide PWM output)
Analog Input Pins	8
DC Current per I/O Pin	40 mA
Flash Memory	32 KB (2 KB for bootloader)
SRAM	2 KB
EEPROM	1 KB
Clock Speed	16 MHz
Dimensions	45 x 18 mm
Weight	7 g

3.2 SG90 Micro Servo Motor

The SG90 is a lightweight micro servo motor used for controlling the spider robot leg joints. Twelve servos are used in total, providing three degrees of freedom per leg (hip, knee, ankle).

Parameter	Value
Model	SG90 Tower Pro
Operating Voltage	4.8V - 6V DC
Stall Torque @ 4.8V	1.8 kg-cm (17 oz-in)
Stall Torque @ 6V	2.2 kg-cm (22 oz-in)
Operating Speed @ 4.8V	0.12 sec/60 degree
Rotation Range	180 degrees
PWM Signal	500-2400 microseconds
Weight	9 g

Dimensions	22.2 x 11.8 x 31 mm

Wire Color Code: Brown (GND), Red (VCC), Orange (Signal/PWM)

3.3 HC-SR04 Ultrasonic Distance Sensor

The HC-SR04 ultrasonic sensor provides non-contact distance measurement for obstacle detection and avoidance during robot navigation.

Parameter	Value
Operating Voltage	5V DC
Operating Current	15 mA
Measuring Range	2 cm - 400 cm
Measuring Accuracy	+/- 3 mm
Ultrasonic Frequency	40 kHz
Effectual Angle	< 15 degrees
Trigger Input Signal	10 microsecond TTL pulse
Dimensions	45 x 20 x 15 mm

3.4 MQ-2 Gas and Smoke Sensor Module

The MQ-2 is a Metal Oxide Semiconductor (MOS) gas sensor that detects smoke and various combustible gases. It operates on the principle of chemisorption.

Parameter	Value
Operating Voltage	5V DC
Power Consumption	~800 mW
Working Current	~150 mA
Detectable Gases	LPG, Smoke, Alcohol, Propane, Hydrogen, Methane, CO
Detection Range	200 - 10,000 ppm
Preheat Time	~20 seconds (for stable readings)
Analog Output (AO)	0.1 - 0.3V proportional to gas
Digital Output (DO)	TTL level based on threshold
Dimensions	32 x 20 x 22 mm

3.5 IR Flame Sensor Module (LM393)

The IR Flame Sensor detects infrared light in the wavelength range emitted by flames. It uses an infrared photodiode and LM393 comparator for signal processing.

Parameter	Value
Operating Voltage	3.3V - 5V DC
Detection Wavelength	760 nm - 1100 nm
Detection Angle	~60 degrees
Detection Distance (lighter)	Up to 80 cm
Comparator IC	LM393
Output Driving Capability	> 15 mA
Sensitivity Adjustment	Onboard potentiometer
PCB Dimensions	32 x 14 mm

3.6 HC-05/HC-06 Bluetooth Module

The Bluetooth module enables wireless communication between the robot and a smartphone or computer for real-time monitoring and control.

Parameter	**Value**
Bluetooth Protocol	Bluetooth V2.0+EDR
Frequency	2.4 GHz ISM band
Operating Voltage	3.3V - 6V
Operating Current	30-40 mA
Communication Range	< 100 m (open space)
Data Transfer Rate	Up to 3 Mbps
Serial Baud Rate (default)	9600 bps (data), 38400 bps (AT)
Interface	UART (TTL level)
Default PIN	1234 or 0000

3.7 LM2596 DC-DC Buck Converter Module

The LM2596 step-down voltage regulator efficiently converts the 7.4V battery voltage to the stable 5V-6V required by the microcontroller and servos.

Parameter	**Value**
Input Voltage Range	4.5V - 40V DC
Output Voltage Range	1.25V - 35V DC (adjustable)
Output Current	3A continuous (max)
Conversion Efficiency	Up to 92%
Switching Frequency	150 kHz
Output Voltage Tolerance	+/- 4%
Protection Features	Thermal shutdown, current limit

3.8 18650 Lithium-Ion Battery

Two 18650 lithium-ion cells connected in series provide the primary power source for the robot.

Parameter	**Value**

Cell Dimensions	18 mm diameter x 65 mm length
Nominal Voltage (per cell)	3.7V
Fully Charged Voltage	4.2V
Cutoff Voltage	2.5V - 3.0V
Capacity Range	1500 mAh - 3500 mAh
2S Configuration Voltage	7.4V nominal, 8.4V fully charged
Cycle Life	300-500 cycles

Safety Warning: Never discharge below 2.5V or charge above 4.2V per cell.

4. BILL OF MATERIALS (BOM)

The following table lists all components required for the construction of the Autonomous Fire and Smoke Detection Spider Robot:

S/N	Component	Qty	Unit Price	Total
1	Arduino Nano V3.0	1	N7,063	N7,063
2	SG90 Micro Servo Motor	12	N2,358	N28,295
3	HC-SR04 Ultrasonic Sensor	1	N2,908	N2,908
4	MQ-2 Gas/Smoke Sensor	1	N4,782	N4,782
5	IR Flame Sensor (5pcs)	1	N5,396	N5,396
6	HC-05/HC-06 Bluetooth	1	N6,449	N6,449
7	LM2596 Buck Converter (5pcs)	1	N6,169	N6,169
8	18650 Battery Holder	2	N1,792	N3,583
9	Rechargeable Li-ion Battery	1	N9,548	N9,548
10	Nano Expansion Shield (2pcs)	2	N4,168	N8,336
11	Jumper Wires Kit (120pcs)	1	N12,415	N12,415
12	Cordless Soldering Iron Kit	1	N13,468	N13,468
13	Screwdriver Set (32-in-1)	1	N5,674	N5,674
14	Miniature Screws (600pcs)	1	N4,665	N4,665
15	Cable Ties, Glue, Tape, Switches	-	-	N13,963
	TOTAL			**N127,585**

Note: Prices reflect Temu order dated December 3, 2025. Actual costs may vary.

5. CIRCUIT DESIGN AND CALCULATIONS

5.1 Power Budget Calculation

A comprehensive power budget analysis ensures the battery and power regulation system can meet the demands of all components.

Component	Quantity	Current Each	Total Current
Arduino Nano	1	50 mA	50 mA
SG90 Servos (idle)	12	10 mA	120 mA
SG90 Servos (moving)	6 active	200 mA	1200 mA
HC-SR04 Sensor	1	15 mA	15 mA
MQ-2 Sensor	1	150 mA	150 mA
IR Flame Sensor	1	20 mA	20 mA
HC-05 Bluetooth	1	40 mA	40 mA
TOTAL (Peak)			**~1595 mA**

Battery Runtime Calculation:
Using 2x 18650 batteries at 2500mAh capacity with average current ~800 mA:
Estimated Runtime = 2500mAh / 800mA = 3.125 hours. With 80% efficiency: ~2.5 hours

5.2 PWM Servo Control Calculation

PWM Frequency: 50 Hz (20 ms period)
Pulse Width Range: 500us (0 degrees) to 2400us (180 degrees)
Center Position: 1500us (90 degrees)
Angular Resolution: (2400-500)/180 = 10.56 us per degree

5.3 Ultrasonic Distance Calculation

Distance (cm) = Time (us) x 0.0343 / 2 = Time (us) / 58
Where Speed of Sound = 343 m/s at 20 degrees C

6. ASSEMBLY PROCEDURES

6.1 Pre-Assembly Checklist

1. Verify all components against the Bill of Materials
2. Test each servo motor individually for proper operation
3. Set LM2596 output voltage to 5V-6V using a multimeter
4. Charge 18650 batteries to full capacity

6.2 Power System Assembly

5. Insert 18650 batteries into the holder ensuring correct polarity
6. Connect battery holder positive (red) wire to LM2596 IN+ terminal
7. Connect battery holder negative (black) wire to LM2596 IN- terminal
8. Adjust LM2596 potentiometer to achieve 5V-6V output

6.3 Spider Leg Assembly (4 Legs x 3 Servos)

9. Mount 3 SG90 servos per leg (hip, knee, ankle joints)
10. Attach servo horns to each servo shaft
11. Connect leg segments using screws through servo horns
12. Route servo wires through frame to central shield

6.4 Sensor Wiring

HC-SR04: VCC to 5V, GND to GND, TRIG to D7, ECHO to D8
MQ-2: VCC to 5V, GND to GND, AO to A0, DO to D4
IR Flame: VCC to 5V, GND to GND, AO to A1, DO to D2
HC-05: VCC to 5V, GND to GND, TXD to D0 (RX), RXD to D1 (TX)

7. SOFTWARE IMPLEMENTATION

7.1 Development Environment

- Arduino IDE 2.x or later
- Required Libraries: Servo.h, SoftwareSerial.h
- Board Selection: Arduino Nano (ATmega328P)

7.2 Core Algorithm

1. Initialize sensors and servos
2. Read MQ-2 smoke sensor value
3. Read IR flame sensor value
4. IF smoke OR flame detected THEN trigger alert via Bluetooth
5. Read ultrasonic distance
6. IF obstacle detected THEN execute avoidance maneuver
7. ELSE continue forward spider walking gait
8. LOOP back to step 2

8. TESTING AND RESULTS

8.1 Component Testing

Component	Test Performed	Expected Result	Status
SG90 Servos	Sweep 0-180 degrees	Smooth rotation	[PASS/FAIL]
HC-SR04	Distance measurement	2-400cm range	[PASS/FAIL]
MQ-2	Smoke detection	Analog value change	[PASS/FAIL]
IR Flame	Flame detection	Digital output LOW	[PASS/FAIL]
HC-05/HC-06	Bluetooth pairing	Successful connection	[PASS/FAIL]
LM2596	Voltage regulation	Stable 5V output	[PASS/FAIL]

8.2 Performance Metrics

- Fire Detection Response Time: [_______] seconds
- Smoke Detection Sensitivity: [_______] ppm threshold
- Walking Speed: [_______] cm/second
- Battery Runtime: [_______] hours

9. CONCLUSION AND RECOMMENDATIONS

9.1 Project Summary

This project successfully demonstrates the design and implementation of an autonomous fire and smoke detection robot using a quadruped spider platform. The integration of MQ-2 smoke sensors, IR flame sensors, and ultrasonic obstacle detection with Bluetooth connectivity creates a mobile safety monitoring system capable of active patrol and early fire warning.

9.2 Key Achievements

- Functional quadruped spider locomotion system with 12 servo motors
- Multi-sensor integration for comprehensive fire/smoke detection
- Wireless communication for remote monitoring and alerts
- Autonomous obstacle avoidance capability

9.3 Future Improvements

- Integration of camera module for visual confirmation
- WiFi connectivity for longer-range communication
- GPS module for outdoor deployment
- Machine learning for improved detection accuracy

10. REFERENCES

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15. Elecfreaks. (2024). HC-SR04 Ultrasonic Ranging Module Datasheet.
16. Handsontec. (2024). MQ-2 Gas Sensor Module Documentation.
17. TowerPro. (2024). SG90 Micro Servo Specifications.
18. ITead Studio. (2024). HC-05 Bluetooth Module Datasheet.

APPENDIX A: PIN CONFIGURATION TABLE

Arduino Pin	Component	Function
D0 (RX)	HC-05 TXD	Bluetooth data receive
D1 (TX)	HC-05 RXD	Bluetooth data transmit
D2	IR Flame DO	Digital flame detection
D3 (PWM)	Servo 1	Leg 1 Hip servo
D4	MQ-2 DO	Digital smoke detection
D5 (PWM)	Servo 2	Leg 1 Knee servo
D6 (PWM)	Servo 3	Leg 1 Ankle servo
D7	HC-SR04 TRIG	Ultrasonic trigger
D8	HC-SR04 ECHO	Ultrasonic echo
D9-D11 (PWM)	Servo 4-6	Leg 2 servos (Hip, Knee, Ankle)
A0	MQ-2 AO	Analog smoke level
A1	IR Flame AO	Analog flame intensity
VIN	LM2596 OUT+	5V regulated power input
GND	Common Ground	System ground reference

Note: Additional servos (7-12) for Legs 3 and 4 connect via servo expansion shield using I2C.

--- END OF DOCUMENTATION ---

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